

15.11.2016 (M)

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Exam

Nov – Dec 2016

Max. Marks:100

Class: B.E.

Name of the Course: Soft Computing

Branch: Comp

Course Code:UCEE705

Duration: 3 hours

Semester: VII

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Max. Marks
Q 1 (a)	Explain Fuzzy Extension principle with the help of an example. Also explain various Fuzzy Relations. OR List various types of Fuzzy Inference systems and explain any one in detail.	10
Q 1 (b)	Design a suitable Fuzzy logic controller for a subway train approaching or leaving a station. The inputs are distance from the station and speed of the train. The output is the amount of brake power applied. Design the Fuzzy logic controller using 4 descriptors for input and output variables.	10
Q2 (a)	Explain Derivative-based Optimization method of Steepest Descent. OR Explain the concept of Simulated Annealing.	10
Q2 (b)	Explain various operators in Genetic Algorithm. Also explain Genetic Algorithm with the help of a suitable example.	10
Q3 (a)	Explain Error Back Propagation Training Algorithm with the help of a flowchart or an example. OR Determine the weights after one iteration for Hebbian learning using bipolar binary activation function with the learning constant is $c=1$, the initial weight vector as $W=[1 \ -1 \ 0 \ 0.5]^T$ and the training input vectors as $X_1=[1 \ -2 \ 1.5 \ 0]^T$, $X_2=[1 \ -0.5 \ -2 \ -1.5]^T$ and $X_3=[0 \ 1 \ -1 \ 1.5]^T$	10

Q3 (b)	What is learning in Neural Networks? Differentiate between Supervised and Unsupervised learning.	10
Q3 (c)	Write short note on Kohonen Self -Organizing Networks.	10
Q4 (a)	Explain Coactive Neuro-Fuzzy Modelling. OR Explain the concept of Neuro Fuzzy spectrum.	10
Q4 (b)	Explain Adaptive Neuro-Fuzzy Inference systems (ANFIS) architecture.	10
Q5	Write short note on Printed Character Recognition. OR Explain how Soft Computing can be used for Color Recipe Prediction.	10